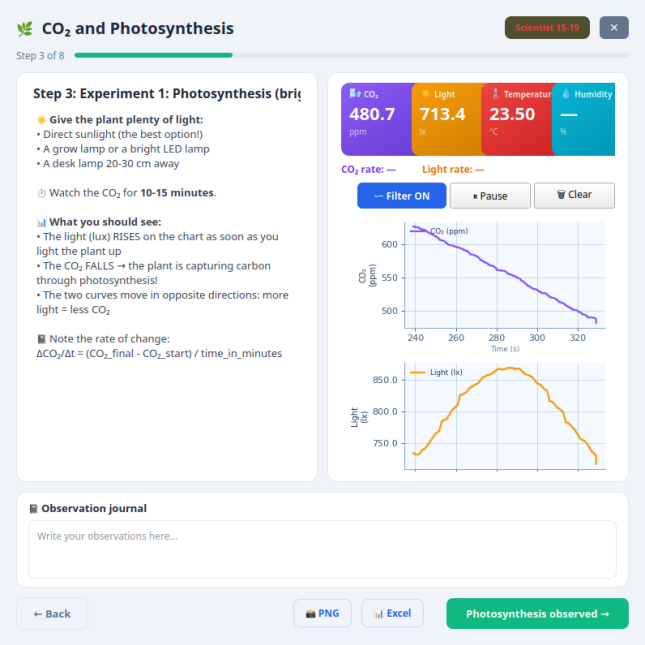
CO_2 and Photosynthesis

A plant in a closed jar — and the point where photosynthesis and respiration cancel exactly

Level: Scientist — ages 15–19 (Romanian school year X) • **Time in class:** 50 min, or 2–3 lessons • **Sensors:** SCD41 (carbon dioxide) and the light sensor (illuminance)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Scientist — upper secondary. The most direct entry is in **year X** (biology: nutrition — photosynthesis and cellular respiration). In **years V–VI**, where photosynthesis and plant respiration first appear, the experiment can be run as a demonstration, without computing gradients. *Limiting factors* and the *light compensation point* are enrichment, suited to keen pupils and to olympiad preparation.

Assumed knowledge

The equations of photosynthesis and cellular respiration. Reading a gradient off a graph and the idea of a rate, in ppm per minute. For the quantitative analysis: an XY chart, a regression line and the idea of a *plateau*, that is, of a factor that limits.

Time in class

Short: 50 min — predictions, theory, photosynthesis in light and respiration in darkness. Full: 2–3 lessons of 50 min each, adding the soil control, the light-intensity series, the compensation point and the limiting factors. **All the app's steps need at least 45 minutes of measuring time alone**; they do not fit into one lesson together with the theory.

Teacher preparation

30 min — **choose and try the plant**, because everything depends on it; assemble the jar and check that it seals; **do a ten-minute trial run** to see whether the CO_2 really does fall in light; and prepare the light source and the shading sheets. The jar is closed only at the start of the first experiment.

Sensor settling time	The SCD41 needs thirty seconds after power-up and gives a new value once every five seconds. After closing the jar or changing the light, **wait about three minutes** (the air mixes, and the sensor has a response time of about a minute), then measure the gradient over **at least three to five minutes**. The gradient counts, not the value.
Sensors and boxes	SCD41 for CO_2 and the light sensor for illuminance; the model in the box is the VEML7700. Both are needed: without measured light the intensity series has no axis. One box per group; a second box allows a **soil control** run simultaneously — the same pot, without the plant. **CO_2 needs recent firmware**: 2.2.0 or later. A box still on the old USB-only firmware (`serial_only`) is recognised, and the app says to flash the WiFi firmware, which works over USB too.

Three routes

Short	50 min: predictions and theory (10), the rig and closing the jar (5), photosynthesis in bright light (13), airing and respiration in darkness (12), computing the net and gross rates and the conclusion (10).
Full	2–3 lessons: lesson 1 as above; lesson 2 the soil control, the light-intensity series and **the compensation point**; lesson 3 the limiting factors, the flux in grams of CO_2 and the link to climate change.
Project	Extension: the same rig with a second jar holding *the same pot without the plant*, as a control, run simultaneously — without it, the soil's respiration adds to the plant's and the two cannot be separated.

1. Where this fits in the curriculum

- **Biology** — in **year X**, plant nutrition: photosynthesis and cellular respiration. The syllabus asks for both processes, and here they are seen *competing*, on the same graph, in the same jar. **That a plant respires by day as well as by night is the commonest misconception at this level**, and the activity demolishes it in one lesson.
- **Biology, years V–VI** — the first encounter with photosynthesis and plant respiration. Here the activity is a demonstration: the teacher handles the jar, and the class watches the line fall in light and rise in darkness, with no gradients and no equations.
- **Enrichment** — *limiting factors* and the *light compensation point* become measurable quantities here, not definitions: pupils find for themselves the intensity at which the CO_2 line goes flat. They suit a biology club or olympiad preparation; check the syllabus in force before treating them as class content.
- **Chemistry** — the activity's two equations are exactly the reverse of each other, which shows very well written one under the other on the board; glucose appears in upper-secondary organic chemistry (check the year in the syllabus in force).
- **Ecology and environmental science** — the carbon cycle, the role of vegetation in the global carbon budget, and the limits of the analogy between a jar and a forest.
- **Mathematics** — the gradient as a rate, in ppm per minute, the regression line as the "honest" gradient of noisy points, and a graph that flattens: a first encounter with the idea of *saturation*.

The core skill is **reading a net rate**. The CO_2 in the jar does not measure photosynthesis; it measures the *difference* between photosynthesis and respiration. The pupil who understands that a flat line does not mean "nothing is happening" but "the two processes cancel" has understood the lesson.

2. Time plan

Route	Time	What happens
Short	**50 min**	Written predictions and the two equations (10), the rig prepared in advance and closing the jar (5), photosynthesis in bright light: three minutes of settling and a five-minute gradient window (13), airing the jar and respiration in darkness (12), the net rate, the gross rate and the conclusion (10).
Lesson 1	50 min	As above, with the predictions written in the app and on the sheet.
Lesson 2	50 min	The soil control (the pot without the plant) and the light-intensity series: maximum, shaded with one sheet, with two, darkness. **The hunt for the compensation point** — the intensity at which the line goes flat. The jar is refreshed with room air whenever the CO_2 falls towards 200 ppm.
Lesson 3	50 min	Limiting factors (light, CO_2, temperature), the flux in grams of CO_2 per hour, the rate-against-intensity graph, and the discussion of the limits of the forest analogy.

The plant decides whether the lesson works

Choose and try the plant before the lesson. A small plant with few leaves in a large jar gives a gradient so small it is lost in the noise; a large plant in a small jar gives a spectacular fall in five minutes — and uses up the CO_2 just as fast.

The rule of thumb: **fill about a third of the jar's volume with leaves** — basil, mint or fresh spinach work very well —, use a 2–5 litre jar and a bright light. **Do a ten-minute trial run** and check one thing only: that the CO_2 really does *fall* in light. If it does not, add leaves or bring the lamp closer before the lesson, not during it.

And water the plant **the day before**, not on the day of the lesson. A thirsty plant closes its stomata, and a plant with closed stomata does not photosynthesise however much light you give it — which is, incidentally, an excellent explanation if it happens. A freshly watered pot, on the other hand, fills the jar with moisture and fogs the walls.

3. Materials

Item	Per group	Notes
MagicBox with SCD41 and light sensor	1–2	Both sensors are needed, and the box needs firmware 2.2.5 or later. The box goes *inside* the jar, so check that it fits before the lesson. The second box is for the control.
Laptop with the app	1	WiFi is far more convenient: a cable coming out of the jar breaks the seal.
Clear jar or box, 2–5 L, with an airtight lid	1–2	**A closed jar only.** With the sensor merely placed next to the leaves in the room's air, nothing reliable can be seen: the air mixes faster than the plant changes it. The second jar is for the **control**.
Fresh green potted plant	1	Basil, mint or spinach. **About a third of the jar's volume**, in leaves. Watered the day before.
An identical pot with the same soil, no plant	1	**The soil control**: the micro-organisms in the soil respire too. Alternatively, cover the plant's soil with plastic film tied round the pot.
Bright light source	1	An LED desk lamp, or a sunny window. **LED is preferable**: it does not heat the jar, and temperature is a limiting factor you do not want to change by accident.
Sheets of white paper	2–3	For shading in steps: one sheet, two sheets, three sheets. The simplest and most repeatable way to reduce the light.
Black cloth or a cardboard box	1	For complete darkness, in the respiration experiment.
Tape or modelling clay	1	For sealing the lid around anything that leaves the jar.
Measuring jug or ruler	1	For the jar's volume (fill it once with water, *without* the box in it) and the pot's volume, for the flux calculation.
Paper towel	1	For wiping condensation off the jar walls between experiments.

Preparation. Choose and try the plant, as the tip above says. Assemble the jar with the box inside and check the seal: close it empty, with the box running, and watch for ten minutes. **In an empty closed jar nothing can change the CO_2, so the line must stay flat**; if it drifts up or down towards the room's value, the jar is not airtight and that must be fixed before the lesson. Measure the jar's volume. Set up the light source at a fixed, marked distance, and cut the shading sheets. Leave the jar *open* until the start of the experiment: a jar closed and lit during the theory uses up its CO_2 before you measure anything.

4. Scientific background for the teacher

The two equations, and why it matters that they are reverses

Photosynthesis:

$$6\,CO_2 + 6\,H_2O \xrightarrow{\text{light}} C_6H_{12}O_6 + 6\,O_2.$$

Cellular respiration:

$$C_6H_{12}O_6 + 6\,O_2 \rightarrow 6\,CO_2 + 6\,H_2O + \text{energy}.$$

As overall equations they are exactly the reverse of each other — and that is the observation the whole activity grows from.

The plant does both, all the time. Cellular respiration never stops: it happens in every living cell, day and night. Photosynthesis happens only in light, in the cells that have chloroplasts. **What the sensor measures is neither process, but their difference**:

$$CO_2 \text{ gradient in the jar} = \text{respiration} - \text{gross photosynthesis}.$$

Hence the names used throughout the activity:

- **Net photosynthesis** is what is measured in light: the fall in CO_2, with its sign changed. It is already photosynthesis *minus* respiration.
- **Respiration** is what is measured in darkness: the rise in CO_2.
- **Gross photosynthesis** is not measured directly; it is computed:

  $$\text{gross photosynthesis} = \text{net photosynthesis} + \text{respiration (measured in darkness)}.$$

  The calculation assumes the plant respires in light as it does in darkness, which is only an approximation — worth saying to able pupils.

The three regimes the class will see. In bright light, photosynthesis exceeds respiration and **the CO_2 falls**. In darkness, photosynthesis is zero, respiration remains, and **the CO_2 rises**. And at a certain intermediate intensity the two match exactly, net photosynthesis is zero, and **the line goes flat**: that is the *light compensation point*.

The compensation point is a real quantity, tabulated in the literature for different species, and far lower than pupils expect — typically 5–50 μmol m^{-2} s^{-1} of photosynthetically active radiation, that is, of the order of a few hundred lux for a shade plant and a thousand or two for a sun plant. **That it can be found experimentally, at school, is the best thing about this activity.**

4.1 Why lux are not the same quantity as photosynthetically active radiation

This is the rig's honest limitation, and it must be said to the class. Plants use **photosynthetically active radiation** (PAR), that is, the number of photons between 400 and 700 nm, measured in μmol m^{-2} s^{-1}. The light sensor measures **illuminance**, in lux, which is radiant power weighted by the sensitivity of the human eye — peaking in the green, exactly where chlorophyll absorbs *least*.

The two quantities can be converted into each other only by a factor that depends on the spectrum of the source: roughly 0.019 μmol m^{-2} s^{-1} per lux for daylight, but noticeably different for a white LED or an incandescent bulb. **Comparisons within a series, with the same lamp, are therefore valid; absolute values in lux cannot be compared with compensation points from the literature.** A good pupil must be able to state that. Moreover, the sensor measures the light where the box sits, not on the leaves.

The light sensor has a limit of its own, which depends on the firmware: with firmware 2.2.5 or

later it changes its sensitivity by itself and reaches about 120 000 lx, but older firmware saturates around 3 800 lx — a nearby lamp or sun through the window can exceed that. If the card sticks around 3 800 lx, flash the new firmware.

4.2 How to measure an honest gradient

The SCD41 has noise of its own of a few ppm and gives one value every five seconds. A gradient computed from two readings — "at the start" and "at the end" — inherits all the noise of those two points. That is why the app computes the gradient as a **regression line** through all the points of a window several minutes long and shows it with its uncertainty, ±. The order of magnitude: over **five minutes**, the gradient comes out to about ±1 ppm/min; over two minutes, to ±3–4 ppm/min, as large as the signal of a small plant. Hence the activity's rule: **the gradient window lasts at least three, preferably five minutes**, and starts only after the three minutes of settling that follow every change.

4.3 Limiting factors, and how to recognise a plateau

The rate of photosynthesis is limited at any moment by *a single* factor — the one in shortest supply. Increasing the others does not help. The three that matter here are:

- **Light.** At low intensities the rate rises almost linearly with illuminance. This is the portion the intensity series measures.
- **CO_2.** In a closed jar the concentration *falls* as the plant works, so after a while **CO_2 itself becomes the limiting factor** and the gradient eases off; left long enough, the line stops at the *CO_2 compensation point*, around 50–100 ppm for most plants. **The restart rule:** when the CO_2 in the jar falls towards 200 ppm, open the lid for a minute, let the room's air in, close it and repeat the step. A gradient measured below 200 ppm tells you about CO_2, not about light.
- **Temperature.** The optimum is usually 25–35 °C. **An incandescent lamp warms the jar and changes this factor without anyone meaning to** — one more reason for LED. The box itself also warms a little when shut in.

The sign that you have run into a limiting factor is the **plateau**: the rate stops rising even though you have raised the intensity. Pupils should look for the plateau, not avoid it.

4.4 What else happens in a closed jar

Moisture. The plant transpires, and in a closed jar the humidity climbs towards 90–95% in ten to twenty minutes. The walls fog, which cuts the light, and water can condense on the box. Watch the humidity card: **above 90%, open the jar, wipe the walls and let the air change** before the next step.

The soil. Potting soil is full of micro-organisms that respire, in light and in darkness. Without a control, their respiration adds to the plant's: net photosynthesis comes out smaller, respiration larger. The control is the same pot with the same soil, without the plant (or the soil covered with film); its gradient is subtracted from all the others.

The sensor's automatic self-calibration. The SCD41 recalibrates itself: it assumes the lowest value it has seen over a few days of running is outdoor air, about 400 ppm. A box that spends hours in a jar with CO_2 below 400 ppm can be fooled, and over the following days it will read *too high*. After a long series of jar work, leave the box running for a while in fresh air, check that it reads 420–450 ppm outdoors and, if not, recalibrate it outdoors (the technical manual shows how).

Pressure. The firmware does not pass the air pressure to the sensor, and the SCD41 works out its ppm for sea-level pressure. Readings are therefore a few per cent too low at altitude

(about 1% for every 80–90 m) and a little off in a warmed jar. For gradients compared with each other the error is negligible; for the flux in grams, correct with the pressure the box measures: $\text{ppm}_{\text{true}} \approx \text{ppm}_{\text{read}} \times 1013.25/P_{\text{hPa}}$.

4.5 From ppm per minute to grams of CO_2

The gradient in ppm/min says how fast the CO_2 *fraction* of the air changes. To find how many grams the plant takes up, you need the number of moles of air in the jar, from the gas law:

$$\frac{\Delta m}{\Delta t} = \frac{\Delta x}{\Delta t} \cdot \frac{P\,V}{R\,T} \cdot M_{\text{CO}_2},$$

where $\Delta x/\Delta t$ is the gradient in ppm/min **multiplied by** 10^{-6} (one ppm is one millionth), P the pressure in pascals, V the volume of air in the jar **in cubic metres** (the jar's volume minus that of the box, the pot and the plant), $R = 8.314$ J/(mol·K), T the temperature in kelvin and $M_{\text{CO}_2} = 44$ g/mol.

Example. A 3 L jar of which the air fills 2.5 L, so $V = 2.5 \times 10^{-3}$ m^3; $P = 101\,000$ Pa; $T = 295$ K; gradient in light -20 ppm/min. Moles of air: $PV/RT = 101\,000 \times 0.0025/(8.314 \times 295) \approx 0.103$ mol. Moles of CO_2 per minute: $20 \times 10^{-6} \times 0.103 \approx 2.1 \times 10^{-6}$ mol/min, that is ≈ 0.09 mg CO_2/min or about 5.4 mg per hour. Dividing by the leaf area (for example 0.02 m^2) gives ≈ 0.27 g CO_2/(m^2·h), a value that can be compared, with care, with those published for leaves (of the order of 1–3 g CO_2/(m^2·h) in bright light). An error of a factor of a thousand (litres instead of cubic metres) is the commonest one.

4.6 The jar and the forest: the limits of the analogy

The app's final step links the activity to climate change, and the link is real — but it is worth setting out correctly. A closed jar is not a forest: it has no living soil with all its cycles, no seasons, and no exchange with the atmosphere. **What transfers is the mechanism, not the numbers.**

What can honestly be said: vegetation absorbs a significant share of annual human CO_2 emissions, and at night and in winter the same vegetation is a net source — exactly the regimes the class has just measured, at the scale of a jar. What *cannot* be said is that planting a given number of trees offsets a particular quantity of emissions; that needs a budget the jar cannot support.

5. Lesson flow

The in-app wizard has **eleven steps**: theory, a prediction, set-up, experiments 1 and 2, a second prediction, experiment 3 (the intensity series), quantitative analysis, limiting factors, the **How sure are we? Slope ± scatter, and a control** step, and conclusion. Before the experiments the app asks for a **written prediction**, keeps it in the group's journal and quotes it back in the conclusion. The short route covers the theory, the set-up and the first two experiments, then skips to the conclusion; the other steps are for lessons 2 and 3.

1. **Written predictions (5 min).** Before anything else: *what will the CO_2 in the jar do in bright light? And in darkness? Is there a light at which nothing happens?* The third question is the important one, and almost nobody answers "yes" before measuring.
2. **Theory (5 min).** The two equations on the board, one under the other, so that it shows they are reverses. Insist on **the net rate**: the sensor sees the difference, not the processes. The jar is still open.
3. **The rig (5 min).** Plant and box in the jar, lid closed and sealed, lamp at the marked distance. Note the time of closing and the starting CO_2.

Technical note — the four cards and what exactly is read off the chart

The sensors. SCD41 for CO_2 (column `CO2`) and VEML7700 for illuminance (column `LUX`). There are four cards on the screen: **CO_2** in `ppm`, **Light** in `lx`, **Temperature** in °C and **Humidity** in per cent. Temperature and humidity are not decoration: **they tell you whether the lamp is warming the jar**, that is, whether you have changed a limiting factor by mistake, and **when the jar needs airing**, before it fogs.

The firmware. CO_2 is read by firmware 2.2.0 or later. A box still on the old USB-only firmware (`serial_only`) may give no CO_2; the app says so and asks for the WiFi firmware, which works over USB too (flashing guide in the `firmware` folder). With firmware older than 2.2.5 the light sensor stops around 3 800 lx.

What is read — the gradient, not the value. The whole activity is done from *gradients*. The absolute CO_2 value in the jar depends on how much air was in it when you closed it and means nothing in itself. Under the cards the app shows **the CO_2 rate as a regression gradient over at most the last five minutes of the step, with ±**: "CO_2 rate (regression over the last 5:00): -12.3 ± 0.9 ppm/min". Until the window covers three minutes, the line says "collecting data (… of at least 3:00), keep the light unchanged". Two warnings appear there by themselves: CO_2 below 200 ppm (the restart rule) and humidity above 90% (wipe the walls).

How long to wait. About **three minutes** after closing the jar and after every change of light, for the air to mix and the sensor to catch up; then the gradient window, of at least three to five minutes. **The restart rule:** if the CO_2 falls towards 200 ppm, open the jar for a minute, close it and repeat the step.

The eleven steps and their buttons. The green button changes its label at every step: **I understand the biochemistry!**, **Setup ready**, **Photosynthesis observed**, **Respiration observed**, **Intensities tested**, **Rates calculated**, **Limiting factors tested**, **Understood** and, at the end, the blue **Finish** button. The two prediction steps end with **My prediction is written**. At the set-up, **Capture** keeps the starting CO_2; at the three experiments it also keeps the window's rate with its ± at that moment, and the app builds its calculations and conclusion from them. In experiment 3 every capture adds a row to the **Light (lx) / CO_2 slope (ppm/min)** table, and the app interpolates the **light compensation point** by itself between the two levels where the gradient changes sign. At the quantitative analysis, type the measured volume of air into the **Volume of air in the jar (L):** field, and the app works out net, respiration, gross, the moles of air (with the box's P and T) and the **Net uptake in the light**, in mg of CO_2 per hour; it does not subtract the soil control, you do. The chart (CO_2 and light) starts at 0:00 at every step, with time in minutes and seconds, so the window's length is read straight off the axis.

Exporting the data. At the measurement steps, **PNG** (all the step's charts) and **Excel** (the step's raw, unfiltered readings, with an `elapsed_s` column of seconds elapsed — exactly what a spreadsheet regression needs) appear. Predictions and notes are saved in the group's journal, in the `logs` folder, and as a **Journal** sheet in the activity's Excel file. Files covering the whole lesson come from the **Data & Export** page, visible at Scientist level too. **Export at the end of every lesson**, because the quantitative analysis is done on gradients computed from the file.

4. **Experiment 1 — photosynthesis, in bright light (13 min).** Lamp at maximum. Three minutes of settling, then a five-minute gradient window; record the rate shown, with its ±, and the illuminance. **The CO_2 falls**: this is *net photosynthesis*. If the CO_2 approaches 200 ppm, apply the restart rule.

5. **Experiment 2 — respiration, in darkness (12 min).** Open the jar for a minute, to start again from the room's air, close it and cover it completely with the black cloth. Three minutes of settling, then the gradient window. **The CO_2 rises**: only respiration remains. The gradient here is *the respiration rate*.

6. **Net rate and gross rate (10 min).** On the board: net photosynthesis = −(the gradient in light); respiration = the gradient in darkness; **gross photosynthesis = net photosynthesis + respiration**. With the soil control, its gradient is first subtracted from both. The conclusion in the app, which shows the pupils their written prediction.

7. **Experiment 3 — the intensity series (lesson 2).** The app first asks for a second prediction: at what light will the line stop? Then: maximum, one sheet, two sheets, three sheets, darkness. At each step, after three minutes of settling, the gradient is measured and **Capture** pressed, which adds the step to the table. **Look for the step at which the gradient is zero, within its ±** — the compensation point, which the app also interpolates. Refresh the jar's air between steps whenever it falls towards 200 ppm, and settle again.

8. **Quantitative analysis (lesson 3).** The graph of net rate against illuminance. The crossing with the horizontal axis is the compensation point; **the value at zero illuminance is the respiration rate itself**, measured in experiment 2, which is a lovely internal check. Then the flux in grams of CO_2 per hour: the pupils work it out by hand and check it against the app's figure once they have typed in the volume of air in the jar.

9. **Limiting factors and the close.** Discuss the plateau at high intensities and connect it to the falling CO_2 in the jar. If you do the app's bonus experiment (breathing towards the plant), do it last: opening the jar ends any series. The **How sure are we?** step gathers the controls: a rate differs from zero only if it clearly exceeds twice its ±; the soil control is subtracted; the series run in both directions must give the same compensation point; the sensor is checked outdoors after a long series. Close with the link — set out correctly — to the global carbon budget.

The investigation, in one table

The question. Is there a light intensity at which the plant neither takes up nor gives out CO_2?
The hypothesis. Written **beforehand**, on the sheet and in the app: yes or no, and if yes, at roughly how many lux. It is not changed afterwards.
What I change. One thing only: **the illuminance**, by adding sheets of paper one at a time.
What I measure. The gradient of the CO_2 line, in ppm/min, as a regression line over at least three to five minutes, with its ±, **and** the illuminance inside the jar at each step.
What I keep the same. The same plant, the same jar, the same lamp distance, the same seal; **the temperature** — which I check on the card at every step, because a lamp that warms changes a different factor from the one I am studying; and **the starting CO_2**, which I bring back to the room's value when it falls towards 200 ppm.
How many repeats. The series is run once in increasing order of shading and, if time allows, once in reverse. **If the compensation point comes out the same both ways, within its ±**, the plant has not tired and the CO_2 in the jar has not fallen far enough to become limiting itself. The soil control is run once, in light, and its gradient is subtracted from every step.
What makes a reading valid — and what makes it an artefact. Valid is a gradient measured after three minutes of settling, over at least three to five minutes, with the temperature unchanged, the jar closed and the CO_2 above 200 ppm. An artefact is anything else: a jar opened in the middle of a window brings back the room's air; **a gradient measured over two minutes is noise**; a gradient below 200 ppm is limited by CO_2, not light; and fogged walls change the light the sensor believes the plant receives.

6. Analysis after the activity

6.1 What the data should look like

Condition	Typical gradient	What it means
Empty closed jar (the seal test)	0 ± 1 ppm/min	**The seal check.** A drift towards the room's value means a leak.
Pot without the plant (the soil control)	0 to +3 ppm/min	The soil's respiration. Subtracted from every other gradient.
Bright light, healthy plant	−5 to −50 ppm/min	Net photosynthesis; depends strongly on how many leaves there are and how big the jar is.
Medium light (one sheet)	−2 to −15 ppm/min	Net photosynthesis still positive.
The compensation point	0 ppm/min, within its ±	The line goes flat. Typically at a few hundred lux.
Complete darkness	+2 to +15 ppm/min	Respiration only. This is the respiration rate.
CO_2 below about 200 ppm, in light	the gradient eases	CO_2 has become the limiting factor: restart with fresh air.
Humidity in the closed jar	climbs towards 90–95% in 10–20 min	The plant's transpiration. Above 90%, air the jar and wipe the walls.
Temperature in the jar, LED lamp	+0 to +2 °C	Acceptable.
Temperature, filament lamp	+5 to +15 °C	**You have changed two factors at once**; the series is not clean.

6.2 From gradients to the compensation point, step by step

1. **Take a gradient for each light step**, with its ±: the app's regression line or, in the spreadsheet, the `SLOPE` function on the `elapsed_s` and `CO2` columns (multiply by 60 for ppm/min). The sign matters: *minus* means the plant is taking up.
2. **Subtract the soil control's gradient**, if you ran it. A soil respiration of +2 ppm/min skews every value in the same direction.
3. **Compute net and gross photosynthesis.** Net = −(the corrected gradient in light); gross = net + the corrected respiration from darkness.
4. **Plot net rate against illuminance**, with ± error bars. Negative points below, positive above, the horizontal axis through zero.
5. **Read off the crossing.** The illuminance at which the curve crosses zero *is* the compensation point. Interpolate between the two neighbouring steps; the error bars say how precisely you found it.
6. **Check internally.** The curve's value at zero illuminance should be roughly the respiration rate measured separately, in darkness. **If the two do not agree, something changed between experiments** — usually the temperature or the starting CO_2.

6.3 Questions to ask afterwards

1. The line went flat at a certain light. Does that mean the plant was doing nothing? Explain what was actually happening in the leaves.

2. What did you measure in light: net or gross photosynthesis? How did you obtain the other one, and what did you have to assume?
3. You measured illuminance in lux, but the plant uses photons between 400 and 700 nm. Why do the comparisons within your series remain valid, and why can the absolute value not be compared with a textbook?
4. When the CO_2 in the jar fell a long way, the gradient eased off even though the light had not changed. Which factor had become limiting, and how did you check?
5. Why does the soil control matter? What exactly could you not rule out without it?
6. Your jar showed that vegetation takes up CO_2 by day and releases it by night. What can you *not* conclude from this about the planet's forests?

6.4 Assessment ideas

- **Low stakes:** the pupil identifies the falling and rising portions on the graph and labels them correctly — photosynthesis dominant, respiration dominant.
- **Standard:** the pupil records the two gradients with their ±, computes net and gross photosynthesis, and explains why the sensor cannot see photosynthesis on its own.
- **Extension:** the full rate-against-illuminance graph, with error bars and the compensation point found by interpolation, the internal check against the respiration rate, the flux in grams of CO_2 per hour, and a written critique of the lux-to-PAR conversion.

Common pitfalls and misconceptions

No control. Without the pot without a plant, the soil's respiration adds to the plant's, and without the empty-jar test any gradient could be a leak. They are the cheapest controls there are, and they change completely what can be claimed.
Gradients measured over two minutes. The SCD41 gives one value every five seconds and has noise of its own; a gradient over two minutes has an uncertainty of ±3–4 ppm/min, as large as a small plant's signal. Three minutes at the very least, preferably five.
A jar left to fall below 200 ppm. The plant is no longer limited by light but by CO_2; the gradient stops responding to shading. Refresh the air and start again.
The wrong name. What is measured in light is *net* photosynthesis. "Gross" is only the sum of net photosynthesis and respiration.
The filament lamp. It warms the jar by ten degrees and changes the temperature — another limiting factor — while you believed you were changing only the light. Use LED and **check the temperature card at every step**.
Opening the jar in the middle of a window. It brings back the room's air and spoils the gradient. The jar is opened only *between* windows, and the settling is then repeated.
The fogged jar. The plant's transpiration brings the humidity close to saturation; the fog cuts the light and can condense on the box. Air the jar above 90% humidity.
"A flat line means nothing is happening." It means the two processes cancel. This is exactly the misconception the activity demolishes.
"Plants only respire at night." The commonest misconception at this level. Cellular respiration never stops; by day it is merely masked by photosynthesis.
The thirsty plant. A plant with closed stomata does not photosynthesise however much light it gets. Water it the day before.
A box that reads too high on the following days. The sensor's self-calibration took the jar's low values for fresh air. Check the box outdoors and recalibrate it if needed.

Safety and classroom management

Lamps get hot, especially filament ones, and here they sit next to a glass jar and to paper for shading. Keep the sheets away from the bulb, do not cover the lamp, and let it cool before putting it away. **Prefer LED** — both for safety and for the cleanness of the experiment.
Glassware. A large jar with an electronic box inside is heavy and slippery. It is held with two hands, stands on a stable surface away from the table edge, and is checked for chips beforehand.
Electronics in the jar. The box is not waterproof: put no water in the jar and do not water the plant while the box is inside. The pot must be dry on the outside, and above 90% humidity the jar is aired so that no water condenses on the box.
Plants and soil. Do not use toxic or irritant plants — spinach, basil and mint are good choices; avoid house plants with irritant sap (for example dieffenbachia or ficus). Nobody tastes anything. Potting soil can carry moulds and bacteria: handle it with gloves or wash hands afterwards, and pupils with pollen or mould allergies keep their distance from the pots.
Cables. If you work over USB, the cable leaves the jar and breaks the seal; WiFi solves both the sealing problem and the trip hazard.

Adaptations for pupils with special educational needs

The graph read, instead of the graph calculated. This is the main adaptation: the pupil does not compute gradients but *describes* the graph — where it falls, where it rises, where it is flat — and puts the right label on each portion. **Recognising the three regimes is the objective itself**, and it is reached without any arithmetic; the rate the app shows can be copied straight across, with no calculation.
Axes drawn in advance. For the rate-against-illuminance graph, the sheet comes with the axes drawn and the horizontal zero line thickened. The pupil places four points and reads off the crossing.
One gradient calculation, fully worked. The first rate comes printed on the sheet, with every stage, as a model. The rest are not pre-computed.
Three real roles for pupils with motor difficulties. The rig means handling a heavy jar, but the roles that matter can be done from a desk:

- **the keeper of conditions** — checks the temperature, humidity and CO_2 cards at every step and has the right to stop the series if the lamp is warming the jar, if it fogs, or if the CO_2 falls towards 200 ppm; this is the role with the greatest scientific value, because it protects the "one factor changed" condition;
- **the timekeeper** — announces the three minutes of settling and the start and end of each gradient window;
- **the secretary** — writes the gradients and their uncertainties as they arrive.

Light sensitivity. The bright lamp can be unpleasant. It can be screened at the side with card so that it lights only the jar, and the roles above done from the opposite side of the table.

If things do not go to plan

The CO_2 does not fall in light. Check, in order: are there enough leaves (about a third of the volume)? is the lamp close enough? is the plant watered? is the jar really closed? **A thirsty plant or too few leaves are the usual causes**, and both are fixed in five minutes.
The CO_2 rises in light and in darkness alike. You have a leak, or the soil respires more than the plant photosynthesises. Run the jar *empty* for ten minutes: if it moves there too, the problem is the seal; if not, run the pot without the plant.

You have no second jar or second box for the control. Run the control *beforehand*, with the same jar and the pot without the plant, for five minutes, and use that gradient as a correction. It is less good than simultaneous, but far better than nothing.
The CO_2 falls towards 200 ppm too quickly. The plant is too big for the jar: remove a few leaves, use a bigger jar, or shorten the window to three minutes and refresh the air more often.
The CO_2 card stays empty. The box probably runs the old USB-only firmware (`serial_only`) older than 1.3.4, and the app says so; reflash it with the current WiFi firmware (2.3.0), which works over USB too.
You cannot find the compensation point. Your steps are probably too far apart. Add sheets one at a time and, if it still does not come out, move the lamp away instead of shading — varying the distance continuously gives finer resolution than the sheets.
You do not have time for three lessons. Do the 50-minute route: photosynthesis and respiration are enough to demolish the central misconception. The compensation point is an extension, not the core.
The box will not connect. Do not stop the lesson. Use the "What the data should look like" table as a printed dataset: pupils compute net and gross photosynthesis, plot the rate-against-illuminance graph from the values given, and find the compensation point by interpolation. **The whole analysis — which is the lesson — is done without the equipment.** Measure again another day, after checking the cable, the port selected in the app and, if you work over WiFi, the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **Royal Society of Biology** — `https://www.rsb.org.uk` — search for photosynthesis and limiting factors; material written for teachers, with tested laboratory variants.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the photosynthesis and the gases simulations; the first lets pupils vary light and CO_2 before they measure them.
- **Nuffield Foundation** — `https://www.nuffieldfoundation.org` — classic practical work on photosynthesis, useful for the sensor-free variants that run alongside.
- **Sensirion** — `https://sensirion.com` — the SCD41 data sheet: response time, automatic self-calibration and pressure compensation.
- **NASA Climate** — `https://climate.nasa.gov` — search for the carbon cycle and the role of vegetation, for the final step, set out correctly.
- **Global Carbon Project** — `https://www.globalcarbonproject.org` — the global carbon budget, if a pupil wants the real numbers behind the climate link.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *what is happening in the leaves when the CO_2 line is flat?* Collected in the last two minutes, it says more about the class's understanding than any mark.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Scientist level the three axes look like this:

- **Measurement practice** — the pupil runs the controls, keeps to the three minutes of settling

and to windows of at least three to five minutes, checks temperature and humidity so as not to change two factors at once, and refreshes the air below 200 ppm.

- **Data interpretation** — the pupil works with *gradients* with their ±, not absolute values, subtracts the control's gradient, tells net from gross photosynthesis, and reads the compensation point by interpolation, not by guessing.
- **Scientific reasoning** — the pupil explains why the sensor measures a net rate, recognises the plateau as the sign of a limiting factor, and states the limitation of the lux-to-PAR conversion.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. Pupil sheet

The sheet the pupils fill in is a separate document, in the folder `docs/en/student_sheets`: the file `co2_and_photosynthesis.pdf`. It has two pages — the predictions, the rig and the basic experiments on the first; the intensity series, the rate-against-illuminance graph and the conclusion on the second —, so it is printed double-sided, one sheet per pupil. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write.

Three things to know before printing it. **The sheet asks for gradients with their ±, not values**: the absolute value in the jar means nothing, and a sheet that asked for it would teach the wrong thing. **The control rows are printed first**, even though some classes will not fill them in — their presence on paper is what makes the question "how do you know it isn't the soil, or a leak?" arise by itself. **The rate-against-illuminance graph comes with its axes drawn and its zero line thickened**, because the compensation point is *the crossing with that line*, and if the line cannot be seen, nobody looks for the crossing. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.